Assuming that the discretization schemes preserve the ergodic property of Langevin dynamics, taking the expectation of these equations and considering the limit as n goes to infinity allows us to compute the invariant average of the schemes, and thus to evaluate the perfect sampling bias introduced by the discretization.

7.4.1 Invariant Averages for Schemes Using the A, B and O Splitting

The analysis of schemes using the splitting strategy coded with the A, B and O pieces is particularly simple, as we use a new random number for each stochastic update meaning that the random numbers and state variables are not correlated between steps. Hence we have

$$\mathbb{E}(\mu_{n,i} q_n) = \mathbb{E}(\mu_{n,i} p_n) = 0,$$

for any i and n. Thus, taking expectations of (7.8)–(7.10) for such schemes, and then considering the limit $n \to \infty$ we obtain a closed, linear system of equations

$$\begin{aligned}
\langle q^2 \rangle_h &= \psi_{11}^2 \langle q^2 \rangle_h + \psi_{12}^2 \langle p^2 \rangle_h + 2\psi_{11}\psi_{12} \langle qp \rangle_h + \bar{\mu}_{11}, \\
\langle p^2 \rangle_h &= \psi_{21}^2 \langle q^2 \rangle_h + \psi_{22}^2 \langle p^2 \rangle_h + 2\psi_{21}\psi_{22} \langle qp \rangle_h + \bar{\mu}_{22}, \\
\langle qp \rangle_h &= \psi_{11}\psi_{21} \langle q^2 \rangle_h + \psi_{12}\psi_{22} \langle p^2 \rangle_h + (\psi_{11}\psi_{22} + \psi_{12}\psi_{21}) \langle qp \rangle_h + \bar{\mu}_{12},
\end{aligned} \tag{7.11}$$

where we have constants

$$\bar{\mu}_{11} = \mathbb{E}(\mu_{n,1}^2), \quad \bar{\mu}_{12} = \mathbb{E}(\mu_{n,1}\mu_{n,2}), \quad \bar{\mu}_{22} = \mathbb{E}(\mu_{n,2}^2).$$

We can solve this linear system to find the invariant averages computed by the chosen discretization scheme for the harmonic oscillator with Hamiltonian (7.6).

Example 7.1 Consider the scheme coded ⟦ABO⟧, where

$$\begin{aligned}
&\llbracket \text{ABO} \rrbracket \\
&\boldsymbol{q}_{n+1} = \boldsymbol{q}_n + h\boldsymbol{M}^{-1}\boldsymbol{p}_n \\
&\boldsymbol{p}_{n+1/2} = \boldsymbol{p}_n - h\nabla U(\boldsymbol{q}_{n+1}) \\
&\boldsymbol{p}_{n+1} = e^{-h\gamma}\boldsymbol{p}_{n+1/2} + \zeta \boldsymbol{M}^{1/2}\mathbf{R}_n
\end{aligned}$$